

AMENDMENT NO. 1 JANUARY 2024
TO
IS 17055 (PART 6) : 2021 EARTH-MOVING MACHINERY — SAFETY
PART 6 REQUIREMENTS FOR DUMPERS
(Second Revision)

(Foreword, para 4) — Substitute the following for the existing:

‘The composition of the Committee responsible for the formulation of this standard is given in Annex C.’

(Page 1, clause 2) — Insert the following new standards in reference clause:

<i>IS No./Other Standards</i>	<i>Title</i>
IS 9000	Basic environmental testing procedures for electronic and electrical items:
Part 3/Sec 3 : 1977	Dry heat test, Section 3 Dry heat test for non-heat dissipating items with gradual change of temperature
Part 5/Sec 1 : 1981	Damp heat (cyclic) test, Section 1 16 + 8 h cycle
IS/IEC 60068	Environmental testing:
Part 2	Tests,
Sec 1 : 2007	Test A: Cold
Sec 14 : 2009	Test N: Change of temperature
IEC 61000-4-5 : 2014	Electromagnetic compatibility (EMC) — Part 4-5: Testing and measurement techniques — Surge immunity test
IEC 61000-4-6 : 2023	Electromagnetic compatibility (EMC) — Part 4-6: Testing and measurement techniques — Immunity to conducted disturbances, induced by radio-frequency fields
JSS 55555	Environmental test method for electronic and electrical equipment

[Page 2, clause 4.1(e)] — Substitute the following for existing:

‘Proximity warning device as per Annex B;’

(Page 5, Annex A) — Insert the following Annex after the Annex A:

‘ANNEX B

(Clause 4.1)

PROXIMITY WARNING DEVICE

A system designed for early detection of static and moving objects, vehicles, human beings encountered within virtual target area during movement of dumper and for triggering warning to the operator to prevent collision or run over. The system shall comprise sensors which may employ one or more or combination of technology/methodology (for detection of objects, vehicles and human beings), control unit(s) [for receiving input from sensor(s), processing it and forwarding necessary input to warning unit] and warning unit to trigger audio visual warning to the operator. The system shall have variable target area during forward movement depending upon speed of the dumper and predetermined target area during rearward movement.

B-1 Minimum Safety Requirements for Proximity Warning Devices

B-1.1 Proximity warning device/system shall be provided for detecting static and moving objects including human beings on its own during the vehicle movement for a specified range, and warn the operator in the operator's station.

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B-1.2 At least one object detection sensor, accessible and not interfering with the dumper's/truck's operation shall be provided both in the front and at rear of the dumper/truck at suitable locations, if it is a standalone/self-contained system. The system shall be ergonomically designed and mounted for operator and maintenance personnel.

B-1.3 The system shall be reliable and be able to provide an adjustable audio-visual warning when it detects static and moving objects including human beings, least height light motor vehicle used in the mine, whichever is less, etc within the virtual target area of respective dumper/truck.

B-1.4 The sensor shall detect static and moving objects in a virtual target area as defined below. The manufacturer shall ensure maximum possible detection coverage in the virtual target area depending on the available fitment area and proximity detection technology defined as below:

B-1.5 Virtual target area in front of dumper/truck:

- a) Width of the virtual target area shall be equal to the width of the dumper/truck plus 0.5 m on both sides;
- b) The inner edges (base line) that represent the width of the virtual target area shall pass through the inner edge of bumper of dumper/truck. The centerline of the virtual target area and the dumper/truck centerline shall coincide; and
- c) Length of the virtual target area shall conform to maximum stopping distance as mentioned in the IS 16479 and the length of virtual target area shall be calculated by the control unit of the system dynamically and automatically with relation to vehicle's speed at any given point of time while covering the blind spot distance observed within the width of virtual target area when the dumper was in static pre-start (ignition – on) condition. While calculating stopping distance in accordance with IS 16479, the test slope percentage may be taken as 6.25 percent (that is, 1 in 16, which is maximum permissible slope of haul road under normal operating conditions). Further, Brake response time and operator response time shall also be considered while calculating the stopping distance.

B-1.6 Virtual target area in rear of dumper:

- a) Width of the virtual target area shall be equal to the width of the dumper plus 0.5 m on both sides;
- b) Length of the virtual target area shall be more than or equal to the length of the dumper;
- c) The inner edges (base line) that represent the width of the virtual target area shall pass through the centerline of the rear axle of the dumper; and
- d) The centerline of the virtual target area and the dumper's centerline shall coincide.

B-1.7 The system shall not detect any obstacles beyond the width of the virtual target area and its detection area shall be restricted along the vehicle's pathway for minimizing false alarm. The system shall have intelligent alert generating mechanisms like indication of obstacle in the vehicle's pathway, whether left, right or center and triggering audio alerts after detection of obstacles, of auto cut-off type to avoid operator inconvenience/distraction. The system shall be provided with provisions for recording of details of warning generated with time stamp with location details (if feasible) to enable easy retrieval and analysis of the immediate past 96 deployment operating hours.

B-1.8 Testing Requirement

The system/components shall comply with the following standards:

- a) IS 9000 : Part 3/Sec 3 — Dry heat test (+ 70 °C);
- b) IS 9000 : Part 5/Sec 1 — Damp heat test (at 40 °C and 95 percent RH for 16 h);
- c) IS/IEC 60068 Part 2/Sec 1 — Cold test (– 10 °C);
- d) IS/IEC 60068 Part 2/Sec 14 — Rapid temperature cycle test;
- e) IEC 61000-4-5 — Surge immunity test;
- f) IEC 61000-4-6 — Immunity to conducted disturbances, induced by radio-frequency fields;
- g) JSS 55555 — Sinusoidal vibration test, frequency: 8 Hz to 500 Hz, acceleration: 40 m/s², duration 2 h in each axis; and
- h) Ingress protection test, IP-66 required.

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B-1.9 Rules/guidelines framed by Ministry of Communications and Information Technology (wireless planning and co-ordination wing), wherever applicable, shall be complied.'

(Page 7, Annex B) — Substitute 'Annex C' *for* 'Annex B' in the title.

(MED 07)

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